

MIDAS GUI — Configuration & Defaults Guide

A step-by-step guide to setting up your own defaults, paths, materials and algorithms.

Focused how-to companion to the main `gui_documentation.md` (§15). Everything is stored in **one per-user JSON config** that overrides the shipped built-in defaults. You can manage it entirely from the **Preferences dialog**, or edit / share the JSON file directly.

1. What you can configure

Without editing any code:

Domain	Examples
Geometry defaults	wavelength λ , pixel size, Lsd, beam centre
Menus	the clickable pixel-size presets and K-edge foils
Materials	your phases for ring simulation (lattice + space group)
Calibrants	the calibrant list on the Calibrate tab
Default paths	default data / calibration / output files & folders
Algorithms	default calibration pipeline, integration kernel, output format, error model, colormap
Visible tabs	which optional tabs are shown (Data Viewer, Mask, Calibrate, Batch are always on)

2. How it works

There is **one per-user config file**. If present, its values override the shipped built-in defaults; if absent, you get the built-ins. There is no separate "group" mechanism — to share settings, export a JSON and hand it to a colleague (§6).

Changes take effect on the next launch (values are read when the window is built). A malformed config file is ignored (built-ins are used) — it will never stop the GUI from starting.

Where the file lives

OS	Path
Linux	<code>~/.config/midas_gui/config.json</code>
macOS	<code>~/Library/Application Support/midas_gui/config.json</code>
Windows	<code>%APPDATA%\midas_gui\config.json</code>

You normally never touch this path by hand — the Preferences dialog reads and writes it. **Settings ▸ Open config folder** jumps straight to it.

3. Method A — the Preferences dialog (recommended)

1. Launch the GUI (`midas-gui` , or `python -m midas_gui`).
2. Open **Settings ▸ Preferences...** (menu bar, top-left).
3. The dialog opens **pre-filled with the full shipped defaults** — you edit from a complete starting point.
Work through the tabs:
 - **Geometry** — wavelength, pixel size, Lsd, beam centre.
 - **Paths** — default files/folders (use the ... button to browse).
 - **Materials** and **Calibrants** — **Add** a row and fill `name`, `a`, `b`, `c`, `α`, `β`, `γ`, `SG` (`SG` = space-group number), **Remove selected**, or edit any cell. Materials appear in the Data Viewer's material dropdown; calibrants in the Calibrate tab list.
 - **Menus** — the pixel-size presets (label + μm) and K-edge foils (element + keV).
 - **Algorithms** — default calibration pipeline, integration kernel, output format, error model, and colormap/theme.
 - **Tabs** — tick which optional tabs are visible. Data Viewer, Mask, Calibrate and Batch Integrate are always on (locked). This setting **applies immediately** on Save — no restart needed (unlike the others).
4. Handy buttons (top of the dialog):
 - **Save current GUI state** — copies the Data Viewer's *current* λ / pixel / Lsd / beam centre into the Geometry fields (set a system up once, then capture it).
 - **Save config to JSON... / Load config (JSON)...** — export or import a config file (how you share settings — see §6).
 - **Reset to shipped defaults** — discards your config and returns to the built-ins.
5. Click **Save as my defaults**. Your per-user file is written; **restart the GUI** to apply.

Because the tables are pre-filled with the shipped list, "remove" and "modify" just work — whatever is in the tables when you Save becomes your complete list. To get the original shipped list back at any time, use **Reset to shipped defaults**.

4. Method B — edit the JSON file by hand

Prefer the dialog for materials/calibrants (it pre-fills the full list). Hand-editing is convenient for a few scalar overrides.

1. Copy the template to your per-user path (§2), e.g. on Linux:

```
mkdir -p ~/.config/midas_gui
cp documentation/config.example.json ~/.config/midas_gui/config.json
```

2. Keep **only the keys you want to change** — every section and key is optional.
3. Save and (re)launch the GUI.

Format

```
{
  "geometry": {
    "wavelength_A": 0.39, "pixel_um": 75.0, "lsd_um": 121000.0, "bc_y": 10.0, "bc_z":
      10.0,
    "pixel_presets": [["GE", 200.0], ["Varex", 150.0], ["Pilatus", 172.0], ["Eiger",
      75.0]],
    "k_edge_foils": [["W", 69.525], ["Au", 80.725], ["Pb", 88.005]]
  },
  "materials": { "Ni (FCC)":
    {"a":3.5238,"b":3.5238,"c":3.5238,"alpha":90,"beta":90,"gamma":90,"sg":225} },
  "calibrants": { "CeO2":
    {"a":5.4116,"b":5.4116,"c":5.4116,"alpha":90,"beta":90,"gamma":90,"sg":225} },
  "paths": {
    "calibrant_tif": "", "calibrant_h5": "", "nickel_h5": "", "nickel_dir": "",
    "nickel_frame0": "", "calib_file": "", "pdf_iq_file": "", "pdf_calib": ""
  },
  "ui": {
    "calibration_pipeline": "one_shot", "integration_kernel": "subpixel2",
    "output_format": "csv", "azimuthal_method": "poisson", "plot_theme": "hot",
    "visible_tabs": ["Calib. Refinement", "Corrections", "PDF Analysis",
      "Texture", "Pump Probe", "Results & Export"]
  }
}
```

Key reference

- **geometry** — numbers in Å / μm / px. `pixel_presets` = [label, size_um] list for the clickable **px** menu; `k_edge_foils` = [element, K-edge keV] list for the clickable **λ** menu.
- **materials / calibrants** — keyed by display name; each is a, b, c (Å), alpha, beta, gamma (deg), sg (space-group number). Paths accept ~ and \$ENV_VARS.
- **paths** — override the default file/folder each tab opens with. Empty "" (or omit) keeps the built-in.
- **ui** — `calibration_pipeline` ∈ one_shot | first_time | four_stage | bayesian | joint; `integration_kernel` ∈ hard | subpixel2 | subpixel4 | polygon; `output_format` ∈ csv | xye | fxye | dat | h5 | 2d_csv; `azimuthal_method` (error model) ∈ poisson | azimuthal | hybrid; `plot_theme` (colormap) ∈ hot | gray | viridis | inferno | plasma | turbo. `visible_tabs` = the list of **optional** tabs to show (Calib. Refinement, Corrections, PDF Analysis, Texture, Pump Probe, Results & Export); the four always-on tabs are implicit. Omit to show all. Unlike the rest, this one applies immediately (no restart).

Important — lists replace. If you include `materials`, `calibrants`, `pixel_presets`, or `k_edge_foils`, that section becomes your **complete** list (it replaces the built-in one). Omit a section to keep the built-in list, or use the Preferences dialog, which pre-fills the full shipped list so you always start complete. Valid JSON only — no comments, no trailing commas.

5. Reset to the shipped defaults

- **In the dialog:** Preferences ▶ **Reset to shipped defaults** (then restart).
- **By hand:** delete your per-user `config.json` (§2) and relaunch.

Either returns every setting to the built-in shipped values.

6. Sharing settings with colleagues

There is no shared-file mechanism to configure — you just move a JSON around:

1. Set everything up via the Preferences dialog.
2. **Preferences ▶ Save config to JSON...** and choose a location (email it, drop it on a share, commit it to a project repo — your choice).
3. A colleague opens **Preferences ▶ Load config (JSON)...**, picks the file, reviews it, and clicks **Save as my defaults**.

(Or simply copy the file into their per-user config path from §2.)

7. Common recipes

- **Set a beamline's default energy and detector:** Preferences ▶ Geometry set `wavelength_A` and `pixel_um` (or click the **px** label → your detector) → Save → **Save config to JSON...** and share it.
- **Add a house sample phase:** Preferences ▶ Materials ▶ Add → lattice + SG.
- **Default to polygon integration + XYE output:** Preferences ▶ Algorithms → kernel `polygon`, output `xye`.
- **Point at your data folders:** Preferences ▶ Paths → set the sample/calibration files & folders.

8. Troubleshooting

- **My change didn't apply.** Restart the GUI (settings load at startup).
- **The GUI ignored my file.** It's probably invalid JSON — the app falls back to built-ins. Re-save through the Preferences dialog (it always writes valid JSON).
- **I lost the shipped materials/calibrants.** A hand-edited file with a partial `materials` (or `calibrants`) list replaces the built-ins. Open Preferences (or **Reset to shipped defaults**) to get the full list back, then re-curate.
- **Back to factory defaults.** Preferences ▶ **Reset to shipped defaults** (or delete the per-user `config.json`).